

Irving R. Barron Martinez

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Current Position

Assistant Professor of Instruction

Jul. 2024-Present

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Department of Electrical and Computer Engineering

Education

Ph.D. in Electrical & Computer Engineering

2024

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

M.S. in Electrical & Computer Engineering

2018

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

M.S. in Electronic Engineering with Specialization in Digital Signal Processing

2015

UNIVERSIDAD AUTÓNOMA DE SAN LUIS POTOSÍ, SAN LUIS POTOSÍ, MEXICO

B.S. in Electronic Engineering

2013

UNIVERSIDAD AUTÓNOMA DE SAN LUIS POTOSÍ, SAN LUIS POTOSÍ, MEXICO

Teaching Experience

Instructor: ECE 487 Strategic Entrepreneurship & Innovation for Engineers

Fall 2025

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Graduate-level course
- Co-developed and Co-teach along with Professor Roberto Colangelo

Instructor: ECE 349 Design Capstone

Spring 2025, 2026

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Undergraduate-level course for seniors
- Co-teach along Dr. Jack Mottley and Daniel Phinney

Instructor: ECE 245/445 Wireless Communications

Spring 2025, 2026

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Undergraduate/graduate-level course

Instructor: ECE 348 Design Seminar*Fall 2024, 2025*

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Undergraduate-level course for seniors
- Co-teach along Dr. Jack Mottley and Daniel Phinney

Instructor: ECE 241 Signals*Fall 2024, 2025*

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

- Undergraduate-level course

Adjunct Professor: Programming*Spring 2016*

UNIVERSIDAD AUTÓNOMA DE SAN LUIS POTOSÍ, SAN LUIS POTOSÍ, MEXICO

- Undergraduate-level course
 - Covered fundamentals of the C++ programming language
- Conducted lectures, developed and graded assignments/exams

Substitute Professor: Linear Algebra*Fall 2015*

UNIVERSIDAD AUTÓNOMA DE SAN LUIS POTOSÍ, SAN LUIS POTOSÍ, MEXICO

- Undergraduate-level course
- Conducted lectures and graded assignments/exams
 - Covered the last quarter of the semester for the absent primary faculty member

Funding

I-Corps: Translation Potential of Enhanced Barcode Technologies*11/2024-10/2026*

\$50,000

- National Science Foundation
- PI: Gaurav Sharma, Co-PI: Irving Barron

Student Course Development Project (SCDP): ECE 241 Signals*Summer 2025*

\$3,960

- University of Rochester Teaching Center
- Hired undergraduate student to develop course content, learning materials, and learning activities
- Six weeks of full-time summer work for course development

Affiliations/Service

AFFILIATIONS**IEEE, Member***2024-Present***IEEE, Graduate Student Member***2021-2024***UNIVERSITY SERVICE**

Chief Technology Advisor at the Ain Center for Entrepreneurship and Innovation

2025-Present

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

Member at the ECE Department Undergraduate Committee

2024-Present

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

Member at the ECE Department Engagement and Enrichment Committee

2024-Present

UNIVERSITY OF ROCHESTER, ROCHESTER, NEW YORK

Presentations

- I. R. Barron** “Dual modulated QR codes: augmented QR codes for carrying a secondary message enabling new applications,” talk at *University of Rochester: Graduate Research Day*, Rochester, New York, 20 Oct. 2023.
- I. R. Barron** and G. Sharma, “Proximally secure communication in public settings using specialized barcodes,” talk at *IS&T Electronic Imaging: Media Watermarking, Security, and Forensics*, Online, 28 Jan. 2021.
- I. R. Barron** and G. Sharma, “Toward CanvasChain: A block chain and craquelure hash based system for authenticating and tracking fine art paintings,” talk at *IS&T Electronic Imaging: Media Watermarking, Security, and Forensics*, San Francisco, California, 29 Jan. 2020.

Publications

PEER-REVIEWED JOURNAL PAPERS

- I. R. Barron** and G. Sharma, “Optimized modulation and coding for dual modulated QR codes,” *IEEE Trans. Image Process.*, vol. 32, pp. 2800–2810, 2023.
- I. R. Barron**, H. S. Yeh, K. Dinesh, and G. Sharma, “Dual modulated QR codes for proximal privacy and security,” *IEEE Trans. Image Process.*, vol. 30, pp. 657–669, 2021.

CONFERENCE PAPERS

- I. R. Barron** and G. Sharma, “Proximally secure communication in public settings using specialized barcodes,” in *IS&T Electronic Imaging: Media Watermarking, Security, and Forensics*, 27 – 28 Jan. 2021, pp. 346.1–6.
- I. R. Barron** and G. Sharma, “Toward CanvasChain: A block chain and craquelure hash based system for authenticating and tracking fine art paintings,” in *IS&T Electronic Imaging: Media Watermarking, Security, and Forensics*, San Francisco, California, 27 – 29 Jan. 2020, pp. 399.1–5.

MAGAZINE ARTICLES

- I. R. Barron** and G. Sharma, “Quashing quishing attacks using self-authenticating dual-modulated QR codes,” in *IEEE Security & Privacy*, pp. 2–10, 2025, doi: 10.1109/MSEC.2025.3530487.

Media

“Solving the Problem of QR Codes,” Allan Timothy

May 2025

US NATIONAL SCIENCE FOUNDATION NEWS

<https://youtube.com/watch?v=rg4tl74CfMg>

“QR Code Security,” Nate Pottker

May 2025

US NATIONAL SCIENCE FOUNDATION DISCOVERY FILES PODCAST

<https://www.youtube.com/watch?v=LC32y3YPqt8&list=PL0ujJTaPsv3cFZCgjHk-XdsD7JjY6wM0t>

“New technology could quash QR code phishing attacks,” Luke Auburn

Feb. 2025

HAJIM SCHOOL OF ENGINEERING & APPLIED SCIENCES, UNIVERSITY OF ROCHESTER

<https://www.rochester.edu/newscenter/qr-code-phishing-definition-quishing-638842/>